

Weekly Lesson Plan – Week 2 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon Week of: August 17, 2026 Course: Engineering Design & Problem Solving (InSPIRESS)				
TEKS / ELPS:				
TEKS: §127.406 engineering design process - identify & define a real problem; teamwork & roles; §127.406 problem definition - requirements, constraints & success criteria; §127.406 generate & evaluate solutions; select & justify a design; §127.406 communicate a design solution; prepare a technical presentation; §127.406 present & defend a design solution; evaluate & give feedback ELPS: c.1A, c.2D, c.3C, c.4G, c.5B				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY AUG 17	TUESDAY AUG 18	WEDNESDAY AUG 19	THURSDAY AUG 20	FRIDAY AUG 21
Students will form engineering firms, choose a real problem, and define why it matters (Design Brief p.1).	Students will define their problem with a stakeholder, evidence, testable requirements, and a success statement (Design Brief p.2).	Students will sketch multiple solutions, choose one against their requirements, and generate an AI product image (Design Brief p.3).	Students will build a 5-beat pitch deck with AI and prepare to defend it under questioning (Design Brief p.4).	Students will student firms will pitch their product and defend it under Q&A from a review panel, scored on a shared rubric.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Students will be assessed by written, group, or oral responses throughout the class period.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
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Name one thing you use every day that is badly designed. What's annoying about it?	Who exactly has your firm's problem, and how do you KNOW it's real?	Why do engineers sketch MORE than one idea before choosing?	What makes you actually believe someone's pitch?	What's one thing a great audience does during a presentation?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
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Step 1: Bell ringer / essential question. Step 2: Bell-work (format + example): name a badly-designed everyday thing (6 min) Step 3: Launch: the firm mission + week map (10 min) Step 4: Worked example: a firm end-to-end (8 min) Step 5: Randomize firms + assign 4 roles (6 min) Step 6: Pick your problem + start Design Brief page 1 (15 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: who has it + how you know (6 min) Step 3: Define like an engineer: who, evidence, requirements, success (12 min) Step 4: Worked example + vocab (10 min) Step 5: Write 3-5 testable requirements - Design Brief page 2 (17 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: why sketch more than one idea (6 min) Step 3: Sketch 3 solutions + choose with a reason/trade-off (10 min) Step 4: Write the product-image prompt + generate (10 min) Step 5: Finish Design Brief page 3 (sketches, choice, image) (19 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: what makes a pitch believable (6 min) Step 3: The 5 beats + AI-build the deck from your packet (10 min) Step 4: Drill the defense: assign who owns each question (12 min) Step 5: Finish deck + one full practice run - Design Brief page 4 (17 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: what a great audience does (6 min) Step 3: Format + rubric (how you are scored) (6 min) Step 4: Design Review: 3-min pitch + 2-min defense per firm (33 min) Step 5: Exit ticket + Week 3 (30 Days of Fusion) tease (5 min)
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state what they accomplished and what worked best.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).				